

A127 · Community Visualization Hub: Integrative Visualization of Multimodal Biomedical Data Across Consortia

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Publicly funded research consortia have generated extensive multimodal biomedical datasets, including genome-scale, single-cell, spatial, and bioimaging assays, yet these resources remain siloed across disparate portals, formats, and schemas, restricting integrative exploration. Community Visualization Hub (CVH), (<https://visualizationhub.org/>) is an open-source web platform that federates these resources, more than 3.75 million files and 165 harmonized metadata tags, into a unified environment for computational biologists, bioinformaticians, and biomedical researchers. Building on Gosling for genome-scale genomics and Vitessce for single-cell, spatial, and imaging data, CVH enables linked exploratory views connecting whole-genome patterns to individual loci, cells, and tissue regions. Users can combine self-generated data with public resources without downloading source files, author Gosling visualizations through a graphical interface, and import and share Vitessce configurations. Authenticated project workspaces allow users to persist, annotate, and share analyses and views internally or with external collaborators. Together, these components enable researchers to compare results across modalities, formulate biologically grounded hypotheses, and share reproducible visualizations with the broader scientific community.